

Bluestock Mutual Fund Analytics Capstone Project

Final Report

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GitHub: [Happy-King-747/bluestock-mf-capstone](https://github.com/Happy-King-747/bluestock-mf-capstone)

Executive Summary

This report presents the findings of a comprehensive data analytics project on Indian mutual fund data conducted as part of the Bluestock Fintech internship programme. The project spans seven days of structured work covering data ingestion, cleaning, exploratory analysis, performance metrics, dashboard development, and advanced risk analytics.

The dataset covers 40 AMFI-registered mutual fund schemes across 10 fund houses, with 64,320 daily NAV records, 32,778 investor transactions, and supplementary data on AUM, SIP inflows, portfolio holdings, and benchmark indices.

Key Outcomes:

- Complete ETL pipeline processing 1,46,834 rows across 10 datasets
- SQLite star schema with 9 tables and 1,11,167 loaded rows
- 15 EDA charts revealing market trends and investor behaviour
- Fund scorecard ranking all 40 schemes on a 0–100 composite score
- Interactive 4-page Power BI dashboard with slicers and drill-through
- Advanced risk metrics — VaR, rolling Sharpe, cohort analysis, HHI
- Standalone fund recommender system (recommender.py)

1. Data Sources

1.1 Provided Datasets

Dataset	Records	Key Fields
fund_master	40	amfi_code, fund_house, category, expense_ratio, risk_grade
nav_history	46,000	amfi_code, date, nav
aum_by_fund_house	90	date, fund_house, aum_crore, aum_lakh_crore
monthly_sip_inflows	48	month, sip_inflow_crore, yoy_growth_pct
category_inflows	144	month, category, net_inflow_crore
industry_folio_count	21	month, total_folios_crore
scheme_performance	40	sharpe_ratio, beta, max_drawdown_pct, aum_crore
investor_transactions	32,778	investor_id, transaction_type, amount_inr, state
portfolio_holdings	322	amfi_code, stock_symbol, weight_pct, sector
benchmark_indices	8,050	date, index_name, close_value

1.2 Live API Data

Live NAV data was fetched from mfapi.in for 6 key schemes using the REST API endpoint: GET https://api.mfapi.in/mf/{scheme_code}. A total of 16,707 historical NAV records were retrieved with retry logic and error handling.

2. ETL Design & Data Quality

2.1 Pipeline Architecture

The pipeline follows a linear flow: Raw data → Ingestion → Cleaning → Database → Analysis → Visualisation → Report. Each step is implemented as a standalone Python script runnable independently or via the master `run_pipeline.py` script.

2.2 Key ETL Decisions

NAV Forward-Fill: Weekend and public holiday NAV values are forward-filled from the last available trading day. This is industry-standard practice — NAV does not change on non-trading days.

CAGR Annualisation: Returns are annualised using 252 trading days per year, not calendar days. This is the SEBI and industry standard for mutual fund reporting.

Duplicate Removal: Composite keys (`amfi_code` + `date`) are used to identify duplicate NAV entries. Zero duplicates were found in the provided data.

Type Conversion: All date columns converted from string to `datetime64`. Numeric columns validated and coerced using `pd.to_numeric(errors='coerce')`.

2.3 Data Quality Summary

Dataset	Rows	Nulls	Duplicates	Issues
<code>fund_master</code>	40	0	0	None
<code>nav_history</code>	46,000 → 64,320	0	0	Forward-filled 18,320 weekend rows
<code>monthly_sip_inflows</code>	48	12	0	<code>yoy_growth_pct</code> null for first 12 months (expected)
<code>scheme_performance</code>	40	0	0	<code>max_drawdown</code> negative — correct financial
<code>investor_transactions</code>	32,778	0	0	None — fully clean
Other 5 datasets	8,627 total	0	0	None

3. Exploratory Data Analysis

3.1 Market Overview

Metric	Value
Total AUM tracked	■ 5,47,000 Cr across 40 schemes
Largest fund	Mirae Asset Emerging Bluechip — ■49,046 Cr
Date range	January 2022 to May 2026 (4+ years)
Folio growth	13.26 Cr (Jan 2022) → 26.12 Cr (Dec 2025) — 97% growth
Total NAV records	64,320 (after forward-fill)
Live NAV records	16,707 from mfapi.in

3.2 Investor Demographics

- 26–35 age group drives 37% of all SIP transactions — the most active investor segment
- Punjab leads all states in total SIP investment volume, followed by Tamil Nadu and Madhya Pradesh
- T30 cities account for 66.72% of transactions vs 33.28% for B30 cities
- UPI is the dominant payment mode, followed by Mandate and Cheque
- KYC compliance: 92% Verified, 8% Pending — uniform across T30 and B30
- Gender split is approximately 60% Male, 40% Female across all transaction types

3.3 Sector Allocation

Sector	Weight %
Banking	18.5%
IT	14.2%
FMCG	11.8%
Pharma	9.3%
Auto	8.7%
Infrastructure	7.4%
Energy	6.9%
Others	23.2%

4. Performance Analytics

4.1 CAGR by Category

Category	Avg 1yr CAGR	Avg 3yr CAGR
Small Cap	24.6%	22.8%
Mid Cap	18.3%	19.2%
Large Cap	12.4%	13.5%
Flexi Cap	15.2%	16.1%
Gilt	5.3%	6.1%
Liquid	7.1%	6.8%

4.2 Risk Metrics Summary

Risk-free rate used: 6.5% annual (RBI repo rate proxy). All Sharpe and Sortino ratios computed using daily returns annualised with $\sqrt{252}$ factor.

Metric	Best Fund	Value
Highest 3yr CAGR	Axis Midcap Regular	35.10%
Best Sharpe Ratio	Mirae Asset Large Cap	1.45
Best Composite Score	UTI Mid Cap Regular	86.86/100
Lowest Expense Ratio	Direct plan funds	0.55%
Best Max Drawdown	Liquid funds	< -1%
Highest Alpha	Equity fund vs index	30.62% annualised

4.3 VaR Analysis (95% confidence)

Category	Daily VaR (95%)	CVaR (95%)	Risk Level
Small Cap	-2.1%	-3.2%	Very High
Mid Cap	-1.8%	-2.7%	High
Large Cap	-1.2%	-1.9%	Moderate
Gilt	-0.3%	-0.5%	Low
Liquid	-0.05%	-0.08%	Very Low

5. Advanced Analytics

5.1 Investor Cohort Analysis

Cohort Year	Investors	Avg SIP	Total Invested	Top Fund
2024	4,456	■ 10,978	■ 15.32 Cr	DSP Top 100 Equity
2025	3,240	■ 11,115	■ 6.40 Cr	SBI Small Cap Fund

5.2 SIP Continuity

Analysis of 1,362 investors with 6+ SIP transactions revealed 97.8% are flagged as at-risk with average gap > 35 days. The 35-day threshold is likely too strict for monthly SIP investors and should be recalibrated to 45 days.

5.3 Fund Recommender

Risk Appetite	Top Fund	Sharpe	3yr Return
Low	ICICI Prudential Liquid	7.68	7.1%
Moderate	ABSL Frontline Equity	1.03	28.96%
High	Nippon India Small Cap	0.94	24.56%

6. Limitations & Recommendations

6.1 Limitations

- Dataset covers only 40 of 1,900+ AMFI schemes — results may not generalise
- Alpha/Beta computed against Index ETF proxy, not actual Nifty 50 returns
- SIP continuity threshold (35 days) flags 97.8% as at-risk — needs recalibration
- Sector HHI uses simulated weights where actual portfolio data is insufficient
- NAV forward-fill means weekend returns are exactly zero, slightly understating volatility

6.2 Recommendations for Investors

- Always choose Direct plans over Regular — 0.5–1% annual saving compounds significantly
- Diversify: Small Cap for growth (accept drawdown), Large Cap for stability, Gilt for hedging
- Maintain SIP consistency — rupee cost averaging requires regular, uninterrupted investment

6.3 Recommendations for Product Development

- Recalibrate SIP at-risk threshold to 45 days based on actual gap distributions
- Target 26–35 age group and Punjab for SIP campaign expansion
- Increase B30 city penetration — 33.28% of transactions from underserved markets
- Build cohort-based nudge system to retain 2025 cohort investors

7. Conclusion

This capstone project successfully demonstrates a complete, production-quality data analytics pipeline on real-world Indian mutual fund data. All seven primary deliverables have been completed along with two bonus deliverables (Streamlit dashboard and fund recommender). The fund scorecard, VaR analysis, and cohort insights provide actionable intelligence for both retail investors and fund house marketing teams. All code is version-controlled on GitHub under the repository [Happy-King-747/bluestock-mf-capstone](https://github.com/Happy-King-747/bluestock-mf-capstone) with a clean commit history reflecting the day-wise progression.